Q'tron Neural Networks for Constraint Satisfaction

Tai-Wen Yue and Mei-Ching Chen
Dept. of Computer Science and Engineering
Tatung University
Taipei, Taiwan, R. O. C.
twyu@mail.cse.ttu.edu.tw and mcchen@mail.ttu.edu.tw

Abstract

This paper proposes the methods to solve the constraint satisfaction problems (CSPs) *using* Q'tron neural networks (NNs). *A Q'tron NN is local-minima free if it is built as a* known-energy system *and is incorporated with the proposed* persistent noise-injection mechanism. *The so-built Q'tron NN, as a result, will settle down if and only if a feasible solution is found. Additionally, such a Q'tron NN is intrinsically* auto-reversible. *This renders the NN operable in a question-answering mode for extracting interested information. A concrete example, i.e., to solve the N-queen problem, will be demonstrated to highlight the main concept.*

1. Introduction

The Hopfield neural network (NN) (Hopfield 1982, Hopfield and Tank 1985) is an important NN model to solve the problems of combinatorial optimization and constraint satisfaction. To solve the problems, the NNs are ingenuously constructed such that the satisfactory solutions[1] are located at the energy valleys of the NNs. The problem solving strategy being workable relies on that the internal energy of a Hopfield NN possesses the monotonically decaying property. This renders the NN being able to settle down rather quickly, and, hence, the problems can be resolved effectively and efficiently. This, however, also reflects that the Hopfield NN, in fact, performs a greedy search. The scheme is appropriate only if each and every energy valley of the NN represents a satisfactory solution to the corresponding problem. Unfortunately, this condition, in general, is not met for combinatorial optimization and constraint satisfaction using Hopfield NNs. This, on another hand, becomes the main pathogeny of the Hopfield NN to usually report poor and, even worse, infeasible solutions (Hopfield 1982).

Several stochastic approaches based on simulated annealing and other techniques have been proposed to deal with the local-minima problem (Aarts and Korst 1989, Tambouratzis 1997). Such attempts have resulted in several alternative neural-network models including Boltzmann (Ackley et al. 1985), Cauchy (Jeong and Park 1989) and Gaussian Machines (Akiyama et al. 1989). They used different probability distributions to generate the strength of random noise, and all of them used a cooling schedule to control the parameter of the underlying distribution. Conceptually, the parameter is considered as the temperature of the system. As the temperature is high, any strength of random noise is likely to be generated. As the temperature is gradually reduced, the probability to generate strong strength of noise, i.e., noise with large magnitude, will be decreased accordingly. As the temperature drops to zero, only zero noise is able to be generated. These reveal that the probability to generate any piece of unbounded noise[2] is always nonzero unless the system temperature has reached zero, and only at this temperature, the convergent phase of the system actually starts. These approaches, in fact, tackle the problem in an 'unknown-energy' fashion. That is, the energy level to qualify the set of all satisfactory solutions is unknown in advance. This is, indeed, the reason that these approaches need unbounded noises.

This paper proposes a general scheme to solve the constraint satisfaction problems (CSPs) (Tsang 1993, Bartak 1998) using the Q'tron neural-network model, which is an extension of Hopfield model. This model of NN features on its noise-injection mechanism to escape from local minima. To build a Q'tron NN for constraint satisfaction, its energy function should be constructed with *known-energy concept* in mind. More specifically, the energy function must be built such that one can tell whether or not a state of the Q'tron NN corresponds to a feasible solution by knowing the energy level of that state. Accordingly, we are able to determine the *bounded noise spectra*[3] for Q'trons in the NN. Furthermore, the random noises for Q'trons will be persistently existent, meaning that none schedule is ever needed to control the distribution of noise strength for Q'trons. In the paper, we will apply the concept to solve a benchmark CSP, say N-queen problem. The problem solving strategy of this novel approach is considerably different from relevant approaches using Hopfield-like NNs, e.g., (Douligeris and Feng 2002, Mandziuk 1995, Tambouratzis 1997).

The organization of this paper is as follows: Section 2 gives a brief overview of the Q'tron NN model. Section 3 describes the general scheme to solve CSPs by building a Q'tron NN as a known-energy system. To manifest the known-energy concept, Section 4 describes two methods to

1 For combinatorial optimization, a satisfactory solution means that the solution is feasible and whose performance is good enough. For constraint satisfaction, it is one that is feasible.

2 A piece of random with extremely large magnitude is considered unbounded.

3 If the maximal magnitude of random noise injected into a Q'tron is bounded, the the Q'tron has a bounded noise spectrum.

Proceedings of the Fourth International Conference on Hybrid Intelligent Systems (HIS'04)
0-7695-2291-2/04 $ 20.00 IEEE
Authorized licensed use limited to: Johns Hopkins University. Downloaded on March 01,2026 at 03:35:01 UTC from IEEE Xplore. Restrictions apply.

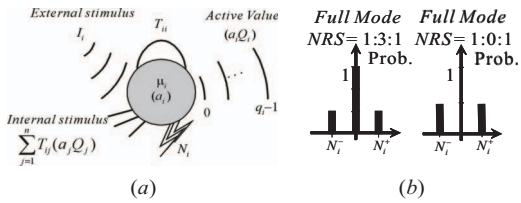


Figure 1. (a) The Q'tron model. (b) Sample noise spectra of Q'tron NNs.

build Q'tron NNs to solve N-queen problem. One is based on the method in Section 3, and the other is by considering the problem nature. Section 5 demonstrates some experimental results. Finally, we will reach a conclusion in Section 6.

2. The Q'tron NN Model

The basic processing element of a Q'tron NN is called a Q'tron, a shorthand of a *quantum neuron*. The number of output-level of each Q'tron can be greater than two. Specifically, let μ_i represent the ith Q'tron in a Q'tron NN, where the output of μ_i, denoted as Q_i, takes up its value in a finite integer set $\{0, 1, ..., q_i - 1\}$ with q_i (≥ 2) being the number of output-level. In addition, Q_i is also weighted by a specific positive value a_i called *active weight*, which stands for the unit excitation strength of that Q'tron. The term a_iQ_i is thus referred to as the *active value* of μ_i, which represents the total excitation strength of that Q'tron. In a Q'tron NN, for a pair of connected Q'trons μ_i and μ_j, there is only one connection strength, i.e. $T_{ij} = T_{ji}$. In Q'tron NN model, each Q'tron in the NN is allowed to be noise-injected. Thus, *noise-injected stimulus* $\hat{\mathcal{H}}_i$ for the Q'tron μ_i is defined as:

$$\hat{\mathcal{H}}_i = \mathcal{H}_i + \mathcal{N}_i = \sum_{j=1}^{n} T_{ij}(a_jQ_j) + I_i + \mathcal{N}_i, \qquad (1)$$

where $\mathcal{H}_i$ denotes the *noise-free net stimulus* of μ_i, which apparently is equal to the sum of *internal stimuli*, namely, $\sum_{j=1}^{n} T_{ij}(a_jQ_j)$, and *external stimulus* I_i. The term $\mathcal{N}_i$ denotes the piece of random noise fed into μ_i, and n denotes the number of Q'trons in the NN. The schematic diagram for a Q'tron is shown in Figure 1(a). In case that $P(\mathcal{N}_i = 0) = 1$, $i = 1, \ldots, n$, the Q'tron NN is said to run in simple mode; otherwise, it is said to run in full mode.

At each time step only one Q'tron is selected for level transition subject to the following rule:

$$Q_i(t+1) = Q_i(t) + \Delta Q_i(t), \qquad (2)$$

with

$$\Delta Q_i(t) = \begin{cases} +1 & \hat{\mathcal{H}}_i(t) > \frac{1}{2}|T_{ii}a_i| \text{ and } Q_i(t) < q_i - 1; \\ -1 & \hat{\mathcal{H}}_i(t) < -\frac{1}{2}|T_{ii}a_i| \text{ and } Q_i(t) > 0; \\ 0 & \text{otherwise,} \end{cases} \qquad (3)$$

where assuming that the ith Q'tron is selected at time $t+1$. Apparently, if we let each Q'tron, say μ_i have $q_i = 2$, $a_i = 1$ and $T_{ii} = 0$, and let the NN run in simple mode, then the Q'tron NN model is reduced to the original Hopfield model (Hopfield and Tank 1985).

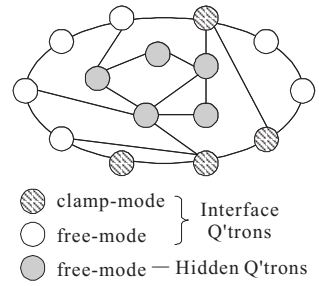


Figure 2. The application model of Q'tron NNs.

2.1. System Energy — Stability

The system energy $\mathcal{E}$ embedded in a Q'tron NN, called *Liapunov energy*, is defined as

$$\mathcal{E} = -\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}(a_iQ_i)T_{ij}(a_jQ_j) - \sum_{i=1}^{n} I_i(a_iQ_i) + K; \quad (4)$$

where n is total number of Q'trons in the NN, and K can be any suitable constant. It was shown that, to run a Q'tron NN in simple mode, the energy $\mathcal{E}$ defined above will monotonically decrease with time (Yue 1992). Therefore, if a problem can be mapped into one which minimizes the function $\mathcal{E}$ given in the above form, then the corresponding NN will autonomously solve the problem after $\mathcal{E}$ reaches a global/local minimum. This reveals that a Q'tron NN performs association, in fact, by 'releasing' its internal energy.

2.2. Known-Energy Systems

A Q'tron NN is called a known-energy system if it can, in probability one sense, settle down onto a state whose energy lying range is known a priori. For example, if a Q'tron NN settles down on a state, say, s, then it is true that

$$0 \leq \mathcal{E}(s) \leq \frac{1}{2}\left(\frac{\Delta}{2}\right)^2, \qquad (5)$$

where $\frac{\Delta}{2} \geq 0$ is a constant (whose value can be tuned), and its reverse must be also true. Then, it is a known-energy system. In (Yue 1992), $\frac{\Delta}{2}$ is called a *solution qualifier*. By tuning the value of $\frac{\Delta}{2}$, we can control the solution quality reported by the Q'tron NN. This is particularly useful for combinatorial optimization since it enables us to approach the global optimum in progressive sense (Yue and Lee 2002). More about these will be discussed in the next section.

2.3. Operating Modes of Q'trons

The instinctive feature of the known-energy system described above allows it to serve as a *question-answering machine* or an *associative memory*. Specifically, each Q'tron

Authorized licensed use limited to: Johns Hopkins University. Downloaded on March 01,2026 at 03:35:01 UTC from IEEE Xplore. Restrictions apply.

can either be operated in *clamp mode*, i.e., its output-level is clamped fixed at a particular level, or in *free mode*, i.e., its output-level is allowed to be updated according to the level transition rule specified in Eq. (2). Furthermore, we categorize Q'trons in a NN into two types: *interface Q'trons* and *hidden Q'trons*, see Figure 2. The former provides an environment to interface with the external world, whereas the latter is functionally necessary to solve certain problems. Hidden Q'trons usually run in free-mode only. Interface Q'trons operated in clamp-mode are used to feed the available or affirmative information (a question) into the NN. The other free-mode interface Q'trons, on the other hand, are used to perform association to 'fill in' the missing or uncertain information (an answer). In this sense, a Q'tron NN, in fact, serves as a question-answering machine or an associative memory. However, to enable a Q'tron NN to perform service in such a manner, it usually must be a *known-energy system*.

2.4. Noise Spectra

Random noise $\mathcal{N}_i$ appearing in Eq. (1) serves to escape the NN 'only' from the local minima representing unsatisfactory solutions. To this end, the strength of $\mathcal{N}_i$ is always bounded, i.e., $\mathcal{N}_i \in [\mathcal{N}_{i-}, \mathcal{N}_{i+}]$. One convenient way for computer simulation is to generate only *bang-bang noise*. In this way, the distribution of noise is specified by the so-called *noise ratio specification* (NRS) of a Q'tron, see Figure 1(b). It is defined as:

$$\text{NRS}_i = P(\mathcal{N}_i = \mathcal{N}_i^-) : P(\mathcal{N}_i = 0) : P(\mathcal{N}_i = \mathcal{N}_i^+) \quad (6)$$

with

$$P(\mathcal{N}_i = \mathcal{N}_i^-) + P(\mathcal{N}_i = 0) + P(\mathcal{N}_i = \mathcal{N}_i^+) = 1; \quad (7)$$

where NRS_i represents the NRS of the ith Q'tron, and $P(\mathcal{N}_i = x)$ represents the probability that the strength of noise $\mathcal{N}_i = x$ is generated. Clearly, if a Q'tron NN runs in *simple mode*, then $NRS_i = 0 : 1 : 0$ for all i.

To make a Q'tron NN as a known-energy system, we let the bounded noise spectra for Q'trons be:

$$\begin{aligned} \mathcal{N}_i &\in [\mathcal{N}_i^-, \mathcal{N}_i^+], \ i = 1, \ldots, n, \text{ with} \\ \mathcal{N}_i^+ &= \max\left(0, \tfrac{1}{2}|T_{ii}a_i| - \tfrac{\Delta}{2}\right) = -\mathcal{N}_i^-. \end{aligned} \quad (8)$$

It was shown that if a Q'tron NN is constructed based on an integer-programming-type energy function, it will be a known-energy system, see (Yue 1992, Yue and Chiang 2002, Yue and Lee 2002). Hence, Eq. (8) will be used to evaluate the bounded noise spectra for Q'trons in sequel.

3. Solving CSPs by Q'tron NNs

The goal of a *constraint satisfaction problem* is to find out solutions conforming to all the constraints of the problem. We will focus on those ones describable using integer programs.

3.1. Standard Integer Programs

Many combinatorial problems can be formulated as integer programming problems in the following standard form

$$\text{minimize} \quad c_1x_1 + \cdots + c_nx_n \quad (9)$$

$$\begin{aligned} \text{subject to} \quad b_{11}x_1 + \cdots + b_{1n}x_n &= d_1 \quad (10) \\ &\vdots \\ b_{m1}x_1 + \cdots + b_{mn}x_n &= d_m \quad (11) \\ x_j \in \{0, \ldots, v_j\}, &\quad j = 1, \ldots, n. \quad (12) \end{aligned}$$

If Eq. (9) is missing, then it is a constraint satisfaction problem. Furthermore, if the constraints of an integer program initially are described in terms of inequalities, they can always be converted into equalities by introducing some *surplus* or *slack* variables (Papadimitriou and Steiglitz 1998).

3.2. Unknown-Energy Systems

We can use a Q'tron NN with n Q'trons to solve the integer problem described in the above form. The number of output levels for each Q'tron can be determined immediately from Eq. (12). That is, let Q_j represent the output level of the jth Q'tron. Using Q_j as the representative of x_j, then we need $Q_j \in \{0, \ldots, v_i\}$, i.e., $q_j = v_j + 1$. Now, consider the following energy function

$$\mathcal{E}_{unknown} = \mathcal{E}_{cost} + \lambda\mathcal{E}_{constraints} \quad (13)$$

with

$$\mathcal{E}_{cost} = \sum_{j=1}^{n} c_jQ_j, \quad (14)$$

$$\mathcal{E}_{constraints} = \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{n}(b_{ij}Q_j - d_i)^2, \quad (15)$$

where $\lambda \gg 0$. It can be easily seen that finding the optimal solution of the integer program is equivalent to minimizing $\mathcal{E}_{unknown}$. However, the minimal value of $\mathcal{E}_{unknown}$, in general, is unknown in advance unless the solution space has been exhausted. The so-built Q'tron NNs are unknown-energy systems because we are not sure whether the energy of a stable state where the NN settles is the lowest or low enough.

3.3. Goal-Directed Energy Function

Now, let's provide a target cost. Let it be G. Then, the goal-directed version of energy function becomes

$$\mathcal{E}_{known} = \mathcal{E}_{diff} + \lambda\mathcal{E}_{constraints} \quad (16)$$

with

$$\mathcal{E}_{diff} = \frac{1}{2}\left(\sum_{j=1}^{n} c_jQ_j - G\right)^2. \quad (17)$$

The minimization of $\mathcal{E}_{known}$ now, in fact, serves to reduce the violation between the solution cost and the target cost. Clearly, $\mathcal{E}_{known} = 0$ if and only if the given goal is reached. The energy function defined in this fashion is called in an integer-programming type in (Yue 1992). Comparing Eq (17) with Eq (15), one can see that the resulting solution cost now becomes another constraint.

To tolerate a certain amount of goal violation, the solutions which satisfy the criterion

$$-\frac{\Delta}{2} \le \sum_{j=1}^{n} c_jQ_j - G \le \frac{\Delta}{2} \quad (18)$$

Authorized licensed use limited to: Johns Hopkins University. Downloaded on March 01,2026 at 03:35:01 UTC from IEEE Xplore. Restrictions apply.
IEEE COMPUTER SOCIETY

now are considered feasible. Then, the energy range corresponding to feasible solutions can then be known exactly. That is

$$0 \leq \mathcal{E}_{known} \leq \frac{1}{2}\left(\frac{\Delta}{2}\right)^2 \tag{19}$$

provided λ in Eq. (16) is positive, and its value is set sufficiently large. Because the value of $\frac{\Delta}{2}$ is correlated with solution quality to solve problems, it is then called a *solution qualifier*. (Yue 1992, Yue and Chiang 2002) described the general method to build Q'tron NNs for combinatorial optimization using integer-programming-type energy functions. There, the method to incorporate with a solution refining scheme to obtain the true optimum for the problems was also sketched.

3.4. The Q'tron NN Construction

For pure constraint satisfaction, i.e., it is not the goal-directed version of optimization problems, the energy function to solve the problems will be

$$\mathcal{E}_{CSP} = \mathcal{E}_{constraints} = \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{n}(b_{ij}Q_j - d_i)^2. \tag{20}$$

Accordingly, the Q'tron NN parameters can be determined as follows:

1. $q_j = v_j + 1$ if $x_i \in \{0, \ldots, v_j\}$, $j = 1, \ldots, n$.
2. $a_j = 1$ for $j = 1, \ldots, n$.
3. $T_{ij}, 1 \leq i, j \leq n$: obtained by mapping Eq. (20) to Eq. (4).
4. $I_j, 1 \leq j \leq n$: obtained by mapping Eq. (20) to Eq. (4).
5. $\mathcal{N}_j^+ = \frac{1}{2}|T_{jj}| = -\mathcal{N}_j^-, 1 \leq j \leq n$: We here have assumed that $\frac{\Delta}{2} = 0$. If $\frac{\Delta}{2} \neq 0$, their values must be determined using Eq. (8).

3.5. The Q'tron NN Operations

The Q'tron NN constructed above usually runs in full mode. To obtain an arbitrary feasible solution, the output levels of all Q'trons are randomly initialized, and all are free. The NN, as a result, will settle down on a feasible state with probability one provided such a state does exist. The detail proof can be found in (Yue and Chiang 2002).

We now describe how the Q'tron NN can be operated in a question-answering manner. Suppose that the NN is in a stable state, and we now want to find a solution with $x_1 = u_1, \ldots, x_m = u_m, m < n$, which is different from the current one. To that, one can clamp the first m Q'trons to output-levels $u_1, \ldots, u_m$, correspondingly, and let all other Q'trons free. Such an operation usually will pull the NN up to a extremely high energy state. Therefore, the free Q'trons will become unstable immediately. Furthermore, due to persistently existent noises, the NN finally is able to settle down to a desirably feasible state with probability one provided such a state does exist.

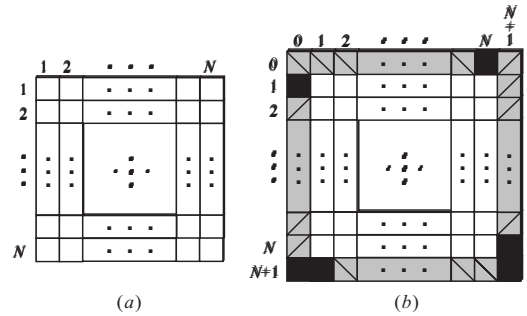


Figure 3. The Q'trons in N-queen Q'tron NN: (a) an $N \times N$ Q'tron plane for interface Q'trons; (b) an $(N+2) \times (N+2)$ Q'tron plane contains also hidden Q'trons; the cells marked black are empty, and the cells marked '/' and '\' are hidden Q'trons to account for northeast and southeast diagonal constraints, respectively.

4. The Q'tron NNs for N-Queen

An N-queen problem is to ingeniously set N queens on an $N \times N$ chessboard so that all of them can live in peace with each other. Only one queen can be in each row, each column, while at most one queen can be in each diagonal. Although N-queen problem is not $\mathcal{NP}$-hard, it is still an attractive problem because it is a scalable problem good for testing algorithm performance and has many applications (Sosic and Gu. 1994), such as VLSI testing, scheduling, resource management, etc.

This section describes two methods to build Q'tron NNs to solve the N-queen problem all based on known-energy concept. One is to formulate the problem as an integer program, while the other is not.

4.1. N-Queen as an Integer Program

We will use an $N \times N$ Q'tron-plane to represent the chessboard of N-queen problem, as was shown in Figure 3(a). Let $Q_{ij} \in \{0, 1\}$, where $i, j \in \{1, \ldots, N\}$, denote the number of queen in cell (i, j), i.e., $Q_{ij} = 0$ represents that a queen is absent in cell (i, j), and $Q_{ij} = 1$ represents that a queen is present in cell (i, j). For convenience, we will directly use Q_{ij}'s, i.e., instead of x_{ij}'s, to formulate the N-queen problem as an integer program. That is,

$$\sum_{j=1}^{N} Q_{ij} = 1, \quad i = 1, \ldots, N \tag{21}$$

$$\sum_{i=1}^{N} Q_{ij} = 1, \quad j = 1, \ldots, N \tag{22}$$

$$\sum_{\substack{i,\, j \,\in\, \{1,\,\ldots,\,N\} \\ i\,+\,j\,=\,m}} Q_{ij} \leq 1, \quad m = 2, \ldots, 2N \tag{23}$$

$$\sum_{\substack{i,\, j \,\in\, \{1,\,\ldots,\,N\} \\ i\,-\,j\,=\,m}} Q_{ij} \leq 1,$$

Authorized licensed use limited to: Johns Hopkins University. Downloaded on March 01,2026 at 03:35:01 UTC from IEEE Xplore. Restrictions apply.
IEEE COMPUTER SOCIETY

$$m = -N+1, \ldots, N-1 \quad (24)$$
$$Q_{ij} \in \{0,1\}, \qquad i,j = 1, \ldots, N. \quad (25)$$

Eq. (22) and (23) require that there is exactly one queen in each row and each column, respectively, and Eq. (24) and Eq. (25) require that there is at most one queen in each of the two directional diagonals.

Because Eq. (24) and (25) are inequalities, the integer program is not in a standard form. For conversion, some slack Q'trons are needed. For conceptually easier, the chessboard is extended to $(N+2) \times (N+2)$, as shown in Figure 3(b). The surrounding Q'trons which are not marked black in the figure are slack Q'trons. These slack Q'trons serve as hidden Q'trons in Figure 2. Accordingly, Eq. (24) to (25) are then replaced as

$$\sum_{\substack{i,j \in \{1,\ldots,N\} \\ i+j=m}} Q_{ij} + Q_{i_m,j_m} = 1, \quad m = 2, \ldots, 2N \quad (26)$$
$$\sum_{\substack{i,j \in \{1,\ldots,N\} \\ i-j=m}} Q_{ij} + Q_{i'_m,j'_m} = 1,$$
$$m = -N+1, \ldots, N-1 \quad (27)$$
$$Q_{ij} \in \{0,1\}, \qquad i,j = 0, \ldots, N+1 \quad (28)$$

where i_m, j_m, i'_m and j'_m are functions for aligning a slack Q'tron to the corresponding diagonal-constraint term, i.e.,

$$(i_m, j_m) = \begin{cases} (m,0), & 2 \le m \le N \\ (m-N-1, N+1), & N+1 \le m \le 2N \end{cases}$$
$$(i'_m, j'_m) = \begin{cases} (0,-m), & -N+1 \le m \le 0 \\ (N+1, N+1-m), & 1 \le m \le N-1 \end{cases}$$

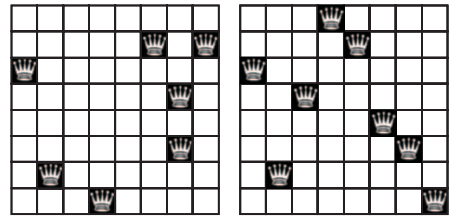
Now, the integer program is in a standard form. Therefore, we can construct the Q'tron NN to solve N-queen problem using the method described in the last section. The so-built Q'tron NN will be denoted as $\mathcal{Q}_1(N)$.

4.2. Another Approach

In this subsection, we construct a known-energy system to solve N-queen problem directly, i.e., without first formulating it as an integer program. The Q'tron NN, as a result, doesn't need any hidden Q'trons. Therefore, the Q'tron NN has the layout of Figure 3(a).

The following four energy sub-terms are to cope with the row constraint, column constraint, northeast diagonal constraint and southeast diagonal constraint, respectively.

$$\mathcal{E}_{-} = \frac{1}{2} \sum_{i=1}^{N} \left(\sum_{j=1}^{N} Q_{ij} - 1 \right)^2 \quad (29)$$
$$\mathcal{E}_{|} = \frac{1}{2} \sum_{j=1}^{N} \left(\sum_{i=1}^{N} Q_{ij} - 1 \right)^2 \quad (30)$$
$$\mathcal{E}_{\backslash} = \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \sum_{k=1}^{N} \sum_{\substack{l=1 \\ k-i=l-j \\ ij \neq kl}}^{N} Q_{ij} Q_{kl} \quad (31)$$
$$\mathcal{E}_{/} = \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \sum_{k=1}^{N} \sum_{\substack{l=1 \\ k-i=j-l \\ ij \neq kl}}^{N} Q_{ij} Q_{kl} \quad (32)$$



(a) (b)

Figure 4. The infeasible local-minima for $\mathcal{Q}_1(8)$ and $\mathcal{Q}_2(8)$.

where $Q_{ij} \in \{0,1\}$ for all $i,j \in \{1,\ldots,N\}$. It can be easily seen that the minima of the above energy sub-terms are all zero, and a feasible state is reached if and only if all these sub-terms are zero. Total energy function to solve the problem is then defined as:

$$\mathcal{E}_{n\text{-}queen} = \mathcal{E}_{-} + \mathcal{E}_{|} + \mathcal{E}_{\backslash} + \mathcal{E}_{/}. \quad (33)$$

It is clear that $\mathcal{E}$ possesses the known-energy property.

Based on Eq. (33), the parameters of the Q'tron NN can be determined using the method described in Section 3.4. This Q'tron NN will be denoted as $\mathcal{Q}_2(N)$.

4.3. Discussions

In the above, we have constructed two Q'tron NNs $\mathcal{Q}_1$ and $\mathcal{Q}_2$ to solve the same problem. Both of them possess known-energy property, and, hence, both are local-minima free. Our empirical studies revealed that the one without hidden Q'trons usually will be computationally more effective, (Yue and Chiang 2001, Yue and Chiang 2004). However, not all problems can be solved in known-energy sense without hidden Q'trons being added. Typical examples were given in (Yue and Chiang 2001, Yue and Lee 2002). From Section 4.1, one then see that reformulating the given problem as an integer program is a good guideline for finding the needed hidden Q'trons.

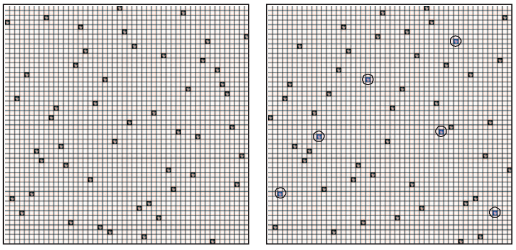
5. Experiments

Figure 4 shows two common stable states for $\mathcal{Q}_1(8)$ and $\mathcal{Q}_2(8)$ running in simple mode, i.e., noise free. As a matter of fact, running in this mode is hardly to obtain a feasible solution especially when N is large. In the following, we demonstrate the experimental results for running $\mathcal{Q}_1$ and $\mathcal{Q}_2$ in full mode.

Some advance experiments revealed that the convergent speed is most effective, both for $\mathcal{Q}_1$ and $\mathcal{Q}_2$, by choosing $NRS = 1 : N : 1$. Therefore, $NRS = 1 : N : 1$ was chosen in the following experiments. Figure 5(a) is a result for $\mathcal{Q}_2(50)$ obtained by randomly initializing the NN and letting all Q'trons free, and Figure 5(b) is obtained in a question-answering fashion. Both of these results were obtained within one minute in a Pentium IV machine.

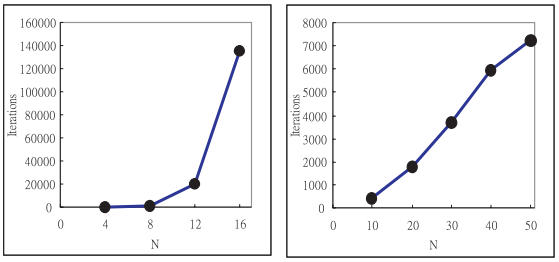
Figure 6(a) and (b) show the average convergent speeds for $\mathcal{Q}_1$ and $\mathcal{Q}_2$ with respect to different N's, respectively. For $\mathcal{Q}_1$, the time needed for convergence seems to exponentially grow with respect to N. For $\mathcal{Q}_2$, it seems linear.

Authorized licensed use limited to: Johns Hopkins University. Downloaded on March 01,2026 at 03:35:01 UTC from IEEE Xplore. Restrictions apply.



(*a*) (*b*)

Figure 5. The results of $\mathcal{Q}_2(50)$ obtained by (a) letting all Q'trons free; (b) running the NN in question-answering mode; the circled Q'trons are clamped to level 1, and all other Q'trons are free.



(*a*) (*b*)

Figure 6. The average number of iterations needed for (a) $\mathcal{Q}_1(N)$; and (b) $\mathcal{Q}_2(N)$ to find a solution.

This is consistent with our empirical studies described previously.

6. Conclusions

This paper proposes the methods to solve the constraint satisfaction problems (CSPs) using the Q'tron NN model. To achieve local-minima free, the NNs to solve the problems must be endowed with known-energy property. This property allows us to determine the bounded noise spectra for Q'trons systematically. Injecting noise into Q'trons persistently based on the spectra renders the NN will never report a infeasible solution, and, hence, the NN can serve as a question-answering machine. To highlight the idea, two methods to built known-energy systems to solve N-queen problem were described. One is based on integer programming, and the other is by considering the problem nature directly. The former method is very general, and it provides a systematical way to find hidden Q'trons needed. However, the latter method is computationally more effective in our experiments. Nevertheless, the Q'tron NNs constructed using both methods are local-minima free. Our experiments also confirmed this. To improve the convergent efficiency for the former type of networks is our important future work.

References

Aarts, E. and Korst, J. (1989), *Simulated Annealing and Boltzmann Machines*, Wiley and Sons, Chichester, England, 1989.

Ackley, D. H., Hinton, G. E. and Sejnowski, T. J. (1985), "A Learning Algorithm for Boltzmann Machine," *Cognitive Science*, vol. 9, pp. 147-169.

Akiyama, Y., Yamashita, A., Kajiura, M. and Aiso, H. (1989), "Combinatorial Optimization with Gaussian Machines," *Proceedings of IEEE International Joint Conference on Neural Networks*, vol. 1, pp. 533-540.

Bartak, R. (1998), On-line Guide to Constraint Programming, http://kti.mff.cuni.cz/ bartak/constraints/, Prague.

Douligeris, Christos and Feng, Gang (2002), "Using Hopfield Networks to Solve Assignment Problem and N-Queen Problem: An Application of Guided Trial and Error Technique," *Proceedings of the Second Hellenic Conference on AI: Methods and Applications of Artificial Intelligence*, vol. 2308, pp. 325-336.

Hopfield, J. J. (1982), "Neural Networks and Physical Systems with Emergent Collective Computational Abilities," *Proceedings of Natl. Acad. Sci. USA*, vol. 79, pp. 2554-2558, April.

Hopfield, J. J. and Tank, D. W. (1985), "Neural Computation of Decisions in Optimization Problems," *Biological Cybernetics*, vol. 52, pp. 141-152.

Jeong, H. and Park, J. H. (1989), "Lower Bounds of Annealing Schedule for Boltzmann and Cauchy Machines," *Proceeding of IEEE International Joint Conference on Neural Networks*, vol. 1, pp. 581-586, 18-22 June.

Mandziuk, J. (1995), "Solving the n-queens problem with a binary Hopfield-type network. Synchronous and asynchronous model," *Biological Cybernetics*, vol. 72, no. 5, pp. 439-446.

Papadimitriou, C. H. and Steiglitz, K. (1998), *Combinatorial Optimization: Algorithms and Complexity*, Prentice-Hall.

Sosic, Rok and Gu., Jun (1994), "Efficient local search with conflict minimization: A case study of the N-queen problem," *IEEE Transactions on Knowledge and Data Engineering*, vol. 6, no. 5, pp. 661-668.

Tambouratzis, T. (1997), "A simulated annealing artificial neural network implementation of the N-queens problem," *International Journal of Intelligent Systems*, vol. 12, no. 10, pp. 739-751.

Tsang, E. (1993), *Foundations of Constraint Satisfaction*, Academic Press, London and San Diego, 1993.

Yue, T. W. (1992), *A Goal-Driven Neural Network Approach for Combinatorial Optimization and Invariant Pattern Recognition*, Phd's thesis, Department of Computer Engineering, National Taiwan University, Taiwan.

Yue, T. W. and Chiang, S. C. (2001), "The General Neural-Network Paradigm for Visual Cryptography," *IWANN*, LNCS 2048, pp. 196-206.

Yue, T. W. and Chiang, S. C. (2002), "Quench, Goal-Matching and Converge The Three-Phase Reasoning Of a Q'tron Neural Network," *Proceedings of the IASTED International Conference on Artificial and Computational Intelligence*, pp. 54-59.

Yue, T. W. and Lee, Z. Z. (2002), "A Goal-Driven Approach for Combinatorial Optimization Using Q'tron Neural Networks," *Proceedings of the IASTED International Conference on Artificial and Computational Intelligence*, pp. 60-65.

Yue, T. W. and Chiang, S. C. (2004), "A Neural-Network Approach for Visual Cryptography Authorization," *International Journal of Neural Systems*, vol. 14, no. 3, pp. 175-187.

Authorized licensed use limited to: Johns Hopkins University. Downloaded on March 01,2026 at 03:35:01 UTC from IEEE Xplore. Restrictions apply.
IEEE COMPUTER SOCIETY